TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Drawing pH–Neutralisation Curves (Strong Acid–Strong Base) and Calculating the Strong Acid's Concentration

The titration drawn as a curve — point-by-point pH calculation, the steep section and the equivalence point, and reading the acid's concentration off the curve's midpoint.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 16 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to plot the pH change while a strong acid is neutralised by a strong base, and to use the curve to determine the acid's concentration. The method: pipette a fixed volume of the acid, add standard NaOH in measured portions, and after each addition either measure the pH (pH meter) or calculate it — in this exercise the values are **calculated** from the un-neutralised H^+ (before equivalence) or the excess OH^- (after it):

Before equivalence: $pH = -\log [(\text{moles } H^+ - \text{moles } OH^- \text{ added}) / \text{total volume}]$

After equivalence: $pH = 14 + \log [(\text{moles } OH^- \text{ added} - \text{moles } H^+) / \text{total volume}]$

2. Model calculation table

25.0 mL of 0.100 M HCl (2.50 mmol H^+) titrated with 0.100 M NaOH:

NaOH added (mL)	Basis	Calculated pH
0	$[H^+] = 0.100$ M	1.00
10.0	1.50 mmol H^+ left in 35.0 mL → 0.0429 M	1.37
20.0	0.50 mmol in 45.0 mL → 0.0111 M	1.95
24.0	0.10 mmol in 49.0 mL → 0.00204 M	2.69
24.9	0.010 mmol in 49.9 mL → 2.0×10^{-4} M	3.70

25.0	**Equivalence — neutral NaCl solution**	**7.00**
25.1	0.010 mmol OH^- excess in 50.1 mL → pOH 3.70	10.30
26.0	0.10 mmol OH^- in 51.0 mL → pOH 2.71	11.29
30.0	0.50 mmol OH^- in 55.0 mL → pOH 2.04	11.96
40.0	1.50 mmol OH^- in 65.0 mL → pOH 1.64	12.36
50.0	2.50 mmol OH^- in 75.0 mL → pOH 1.48	12.52

Two features carry the whole meaning of the curve: the pH moves **slowly** while either reactant is in comfortable excess, then swings through **nearly 7 pH units (3.7 → 10.3) for 0.2 mL** around equivalence — the **steep section** — and the equivalence point itself sits exactly at **pH 7**, because NaCl does not hydrolyse (Chemistry-I Topic 3).

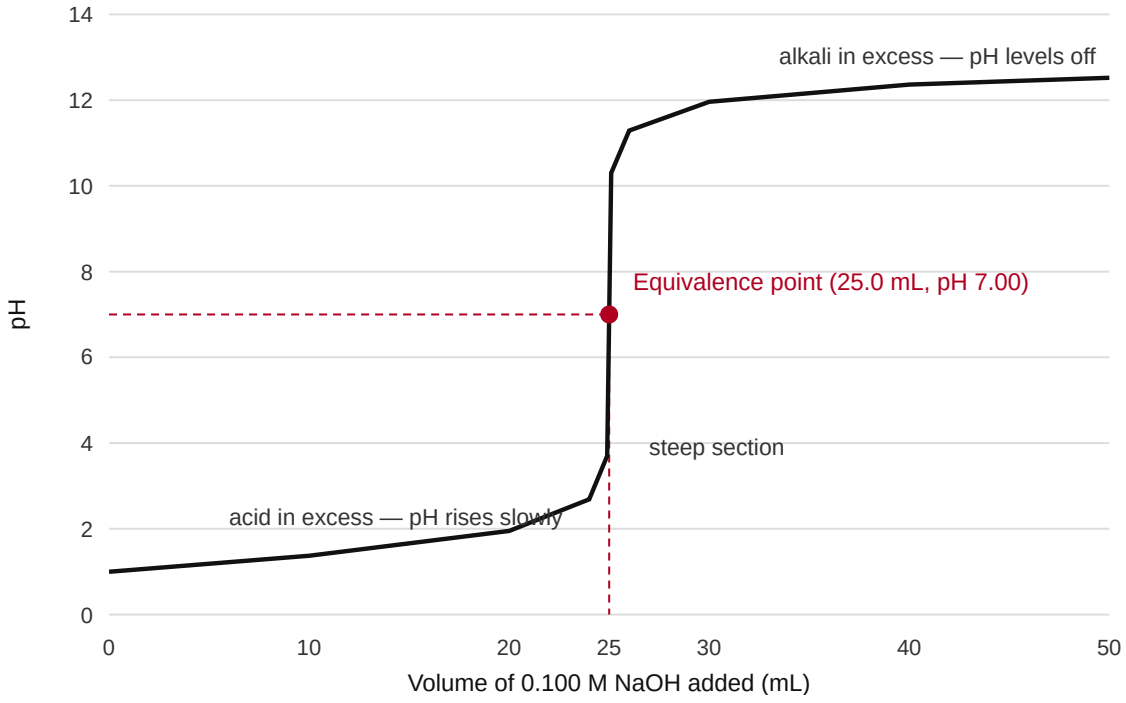


Figure 1. Calculated pH curve for 25.0 mL of 0.100 M HCl titrated with 0.100 M NaOH. The near-vertical section is centred on the equivalence point (25.0 mL, pH 7.00).

3. Determining the acid’s concentration from the curve

WORKED EXAMPLE — AN UNKNOWN ACID

25.0 mL of an HCl solution of unknown concentration is titrated with standard 0.100 M NaOH and its curve plotted as above. The steep section’s midpoint (the equivalence point) falls at **22.5**

mL.

Moles of NaOH at equivalence = 0.100 × 0.0225 = 2.25 × 10^{-3} mol = moles of HCl in the pipette (1 : 1).

Concentration of the acid = 2.25 × 10^{-3} / 0.0250 = **0.0900 M**.

The curve contributes what a single endpoint cannot: the whole shape verifies that the acid is strong (steep section through pH 7) and locates the endpoint as the midpoint of that section — more robust than any one indicator's colour change.

4. Indicator choice from the curve

Any indicator whose range falls inside the steep section (≈ pH 3.7–10.3 here) changes within a drop or two of equivalence: **methyl orange and phenolphthalein both serve** for strong–strong titrations — the conclusion Experiments 14–15 used. The curve also shows why the same freedom does not exist for weak acids or bases: their steep sections are shorter and displaced, and the indicator must be matched to them (Topic 3).

5. Precautions and practical notes

- Plot calculated points from *total* volume — forgetting the added titrant volume is the classic arithmetic slip, worst near the start.
- With a pH meter, standardise it against buffer solutions (pH 4, 7, 10) before use, rinse the electrode between readings, and add titrant in small portions near equivalence where the pH races.
- Read the equivalence volume at the **midpoint of the steep section** (or the maximum slope), not at the first point that reaches pH 7 by eye.
- Temperature affects pH slightly (Kw varies); note the laboratory temperature with meter work.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Calculate the pH after 15.0 mL of the 0.100 M NaOH has been added in the model titration.
2. Calculate the pH after 35.0 mL has been added.
3. An unknown strong acid (25.0 mL) gives an equivalence volume of 18.0 mL with 0.100 M NaOH. Find its concentration.
4. Why is the equivalence point exactly pH 7 in this curve, and when would it not be?
5. Explain, from the table, why one drop near 25 mL changes the pH by several units while 10 mL at the start changes it by 0.4.

6. State the indicator conclusion the curve justifies, and the condition an indicator must satisfy.

Answer key

1. H^+ left = 2.50 − 1.50 = 1.00 mmol in 40.0 mL → 0.0250 M → pH = 1.60.
2. OH^- excess = 3.50 − 2.50 = 1.00 mmol in 60.0 mL → 0.0167 M → pOH 1.78 → pH = 12.22.
3. Moles NaOH = 0.100 × 0.0180 = 1.80×10^{-3} = moles acid; concentration = $1.80 \times 10^{-3}/0.0250$ = 0.0720 M.
4. Because the salt formed, NaCl, is the salt of a strong acid and a strong base: neither ion hydrolyses, so $[H^+] = [OH^-] = 10^{-7}$ at equivalence. It would not be 7 if either partner were weak — the salt would hydrolyse and the equivalence pH would shift alkaline (weak acid) or acidic (weak base).
5. Early on, the added alkali only trims a large excess of H^+, and pH being logarithmic, trimming 40% of 2.5 mmol changes it little. Near equivalence almost no H^+ remains, so a single drop removes a large *fraction* of what is left — and then overshoots into excess OH^- — carrying $[H^+]$ through many powers of ten within a fraction of a millilitre.
6. Both methyl orange and phenolphthalein are valid here. The condition: the indicator's colour-change range must lie within the curve's steep section, so the colour change and the equivalence point coincide to within a drop.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib